

Artificial Intelligence: An Overview

Artificial intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and learn. The term was coined by John McCarthy in 1956 at the Dartmouth Conference, which is widely considered the birth of AI as a field.

1. Machine Learning

Machine learning (ML) is a subset of AI that enables systems to learn from data without being explicitly programmed. Algorithms identify patterns, make decisions, and improve their performance with experience. Supervised learning, unsupervised learning, and reinforcement learning are the three main paradigms.

Deep learning uses multi-layered neural networks to model complex patterns. Convolutional neural networks (CNNs) excel at image recognition, while recurrent networks (RNNs) handle sequential data such as text and speech.

2. Natural Language Processing

NLP enables computers to understand, interpret, and generate human language. Applications include chatbots, sentiment analysis, machine translation, and text summarisation.

Transformer-based models like BERT and GPT have revolutionised the field since 2017.

Large language models (LLMs) are trained on massive text corpora and can perform a wide variety of language tasks with minimal fine-tuning, a property known as few-shot learning.

3. Computer Vision

Computer vision allows machines to interpret visual information from the world. Use cases include autonomous driving, medical image analysis, facial recognition, and quality control in manufacturing. Object detection frameworks such as YOLO and Faster R-CNN achieve near-human accuracy on benchmark datasets.

4. Ethics in AI

As AI becomes more prevalent, ethical considerations grow in importance. Key issues include algorithmic bias, data privacy, transparency, and accountability. Regulatory bodies around the world are developing frameworks to ensure AI is developed and deployed responsibly.

5. Future of AI

Research frontiers include artificial general intelligence (AGI), which would match human cognitive ability across any domain. Neuro-symbolic AI aims to combine the pattern-recognition strength of neural networks with the reasoning power of classical AI. Quantum computing may accelerate training of future AI models.